Natural Language Processing

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Chapter 1

Introduction

الرحمن، The Most Beneficent

علم القرآن،

Has taught (you mankind) the Qur'an

خلق الإنسان، He created man

علمه البيان He taught him

eloquent speech (Qur’an, 55:1-3)

(Eloquence) البيان

- Language (Expressiveness)
- Thinking and Reasoning
- Communication (skill)
  - Be(ing)

NLP

- Goal of NLP: get computers to perform useful tasks involving human language, e.g.:
  - enabling human-machine communication
  - improving human-human communication
  - doing useful processing of text or speech, etc.
- Conversational agents (e.g. HAL 9000 computer in Stanley Kubrick's film 2001*: A Space Odyssey; Apple SIRI, Amazon Alexa, …*)
  - Analyse language input (**Automatic Speech Recognition** and **N.L. Understanding**)
  - Analyse language output (**NL generation** and **Speech Synthesis**).
- Machine Translation

NLP (cont.)

- Web-based Question Answering
  - **Definition** questions
  - **Factoid** questions.
  - More complex questions (**How**, **Why**):
    - information embedded in other text on a Web page,
    - doing **inference** (drawing conclusions based on known facts),
    - synthesizing and summarizing information from multiple sources or web pages.
- Modern understanding systems, including **Info. Extraction**, **Word Sense Disambiguation**
- This chapter summarizes the kinds of knowledge necessary for these tasks (and others like **spell correction**, **grammar checking**, …) and (throughout the course) the needed mathematical models.

NLP

Questions:

1. What is required for computers to learn useful and interesting tasks that use human language?

2. What can we learn from this effort regarding human language processing?

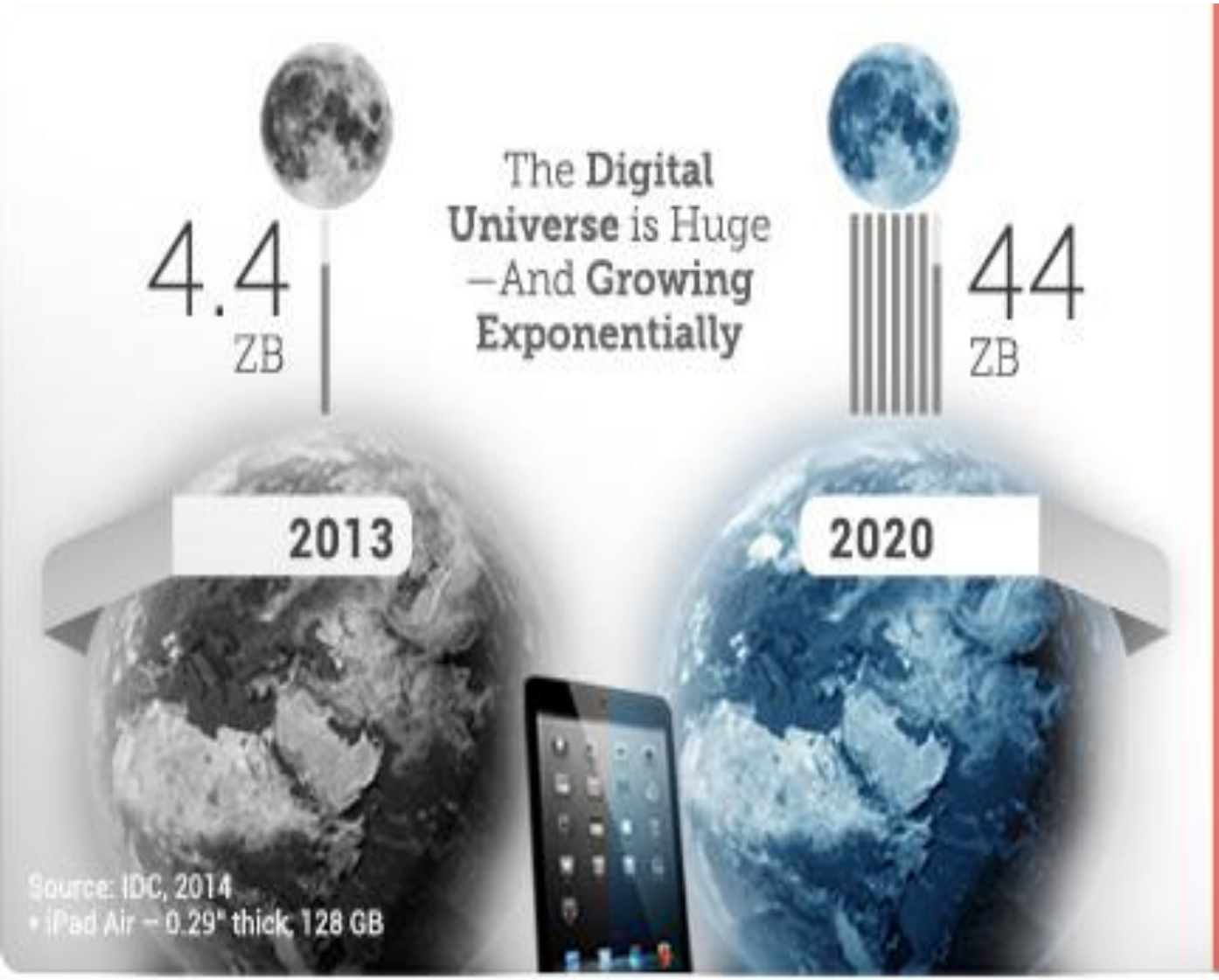
Is NLP Important?

- A huge amount of knowledge now available electronically in the form of NL texts (online journals, Wikipedia, etc.)
  - In 2025, global data creation projected to grow to about 180 zettabytes (10^{21} Bytes)
- Conversational agents now an important form of human-machine communication and work: think of **chatbots**, **generative AI**
- Much human-human communication is facilitated by machines: e.g. **MT** and **Speech-to-speech MT**, **text summarisation**, etc.

"Breaking Down The Numbers: How Much Data Does The World Create Daily in 2024?" (Edge Delta)

- **Facebook** produces ~ 4,000 TB daily, ranking #1 in "most visited sites globally in 2023".
- **X (Twitter)** garners around 500 million tweets daily, which amounts to 560 GB of data.
- **Google** processes around **3.5 billion searches daily**, amounting to **20 PB**.
- In **2023**, the average **internet user** created about 1.7 MB of data per second, i.e. **146.88 GB daily**.

The Digital Universe (7th EMC Digital Universe study, IDC, 2014)



If the Digital Universe were represented by the memory in a stack of tablets, in **2013** it would have stretched two-thirds the way to the Moon*

By **2020**, there would be 6.6 stacks from the Earth to the Moon*

An Information Tsunami!

- 5.44 billion users on the internet (as of S1-2024)
- One person downloading the entire 2025 Global Datasphere at an average of 25 Mb/s would take 1.8 billion years,

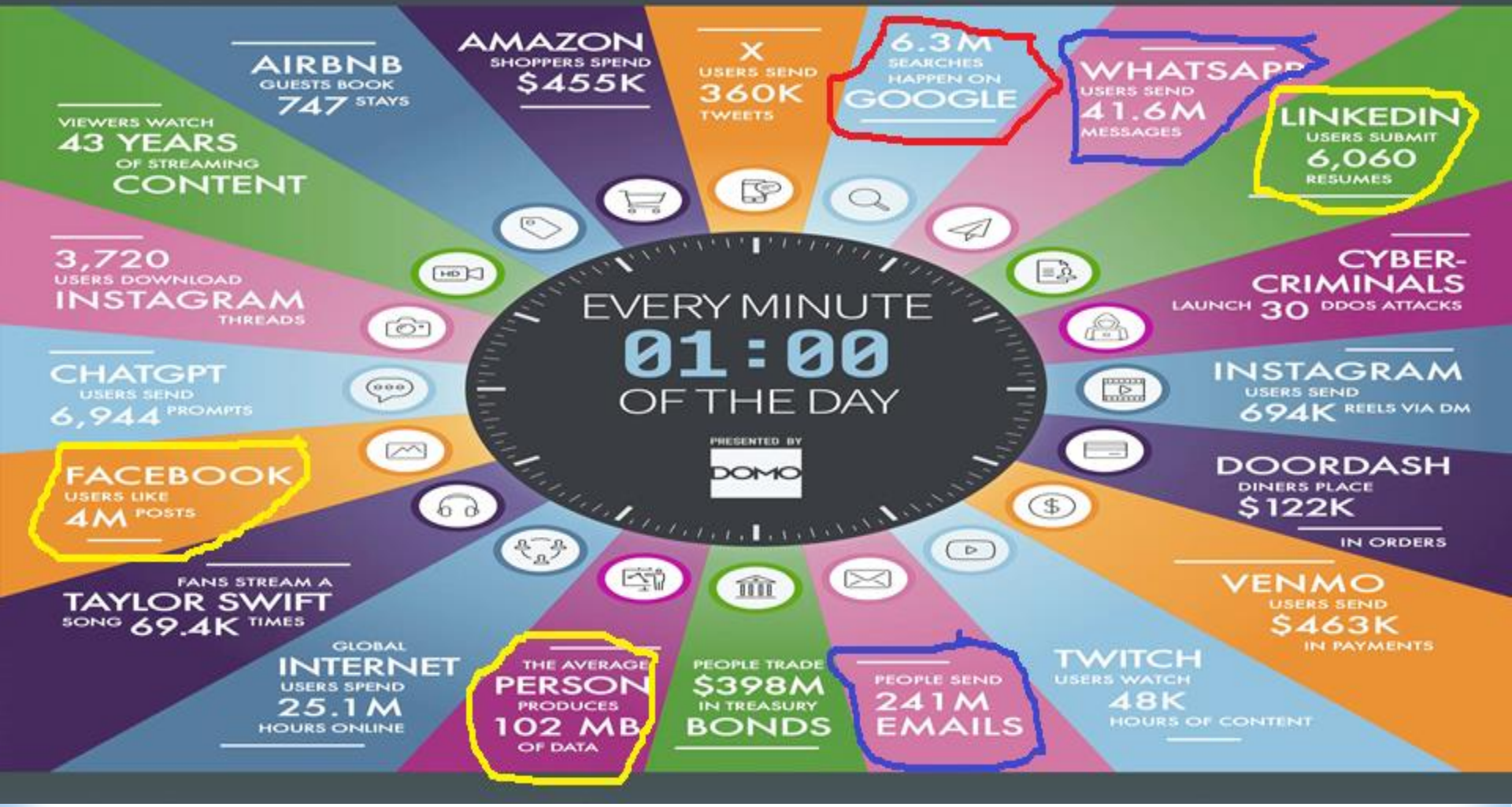
OR for the world population 81 days (working 24x7)

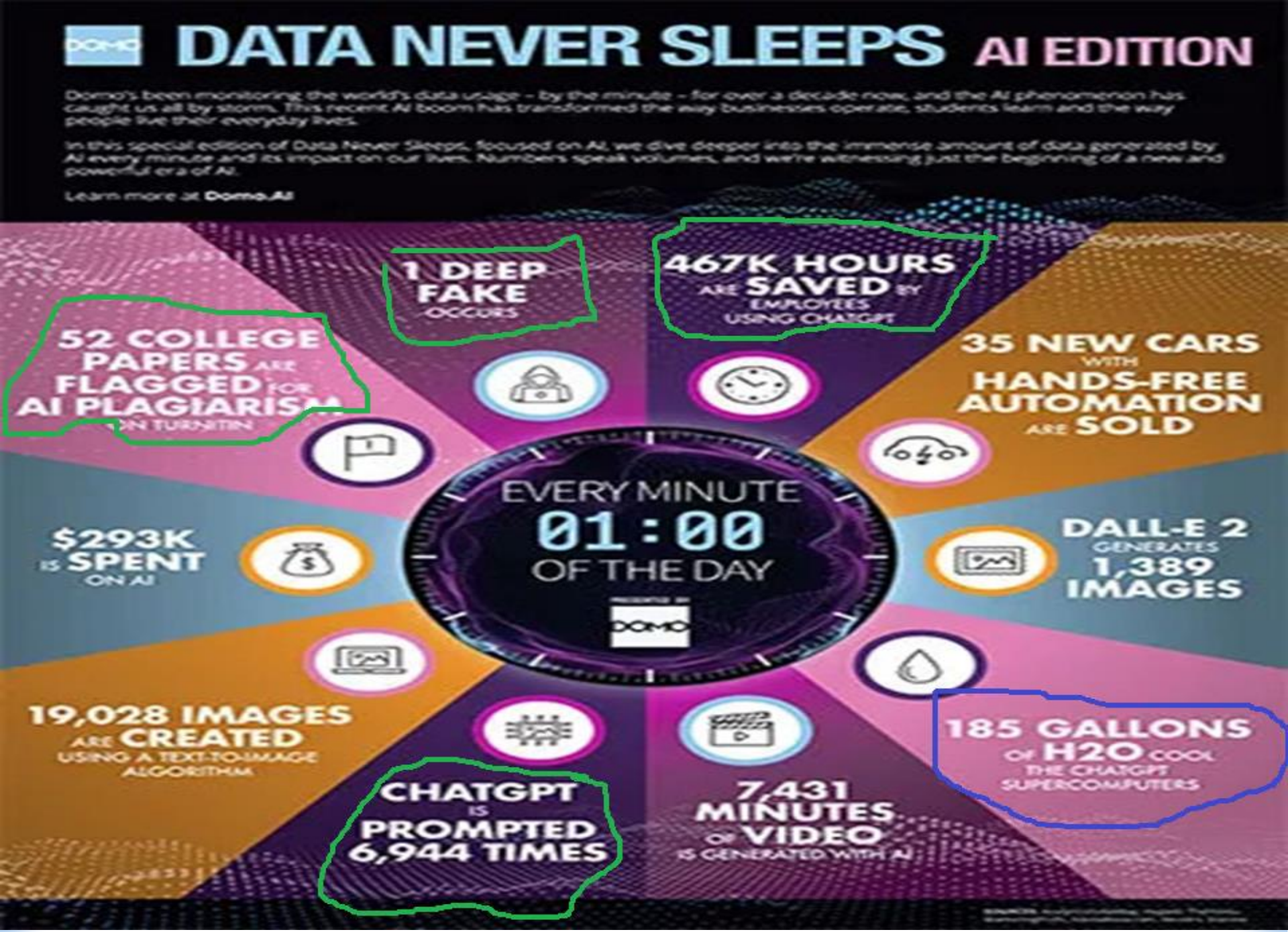
- Using big data, Netflix saves $1 billion per year on customer retention (195.15 M customers as of 3rd Quarter 2020) on its 100 million customers (https://insidebigdata.com/2018/01/20/netflix-uses-big-data-drive-success/)

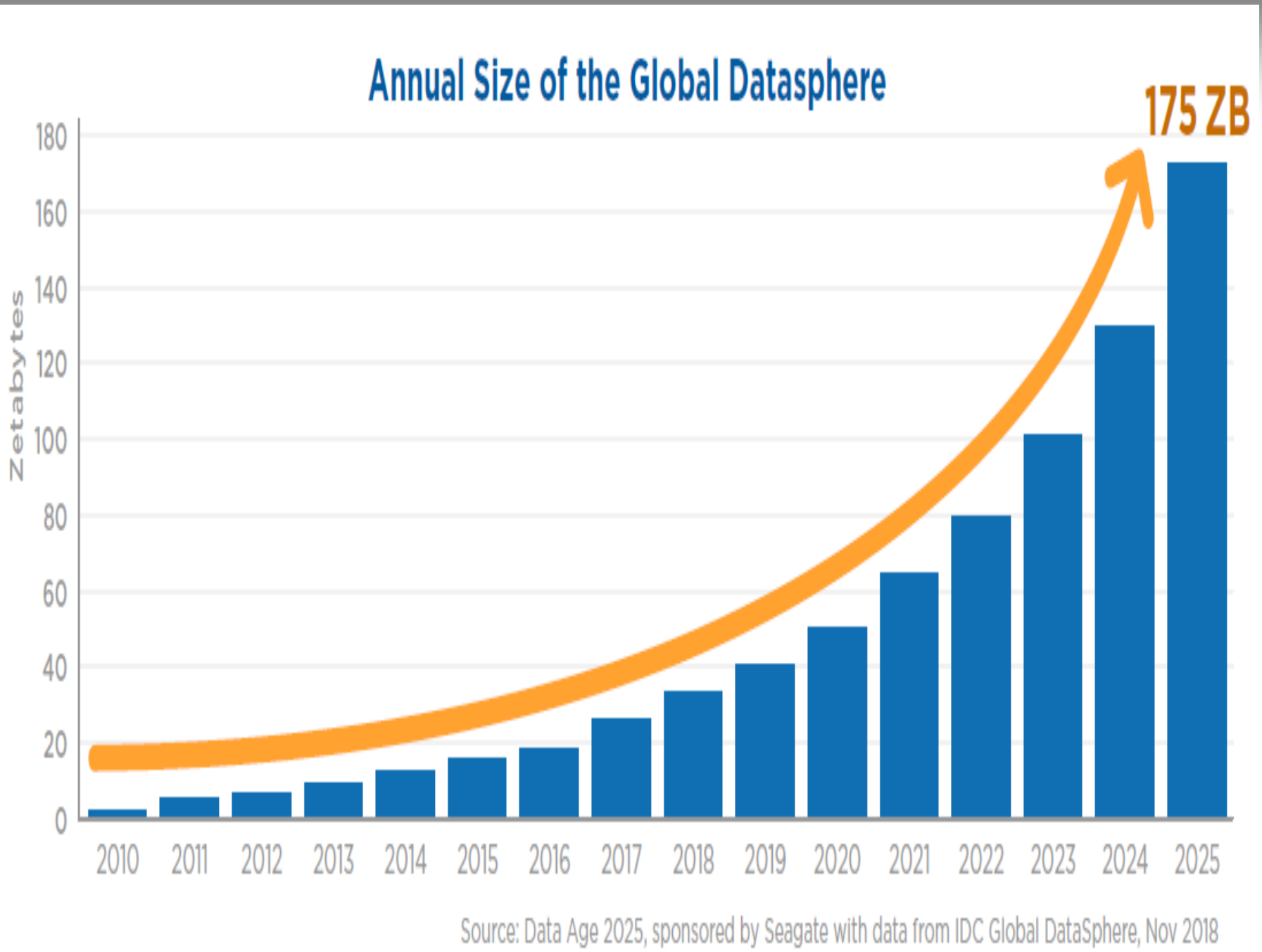
Data Never Sleeps 11.0

Domo has been keeping tabs on the world's data usage—in a minute—for over a decade now. What the numbers consistently show is that how we use data is always evolving—and that data isn't slowing down. We're also seeing some big changes. The rise of Artificial Intelligence (AI) is reshaping the way we communicate, work, and create. Digital payments continue to replace traditional transactions. Taylor Swift streams in countless headphones. And a rash of cybercrime grows alongside these digital experiences.

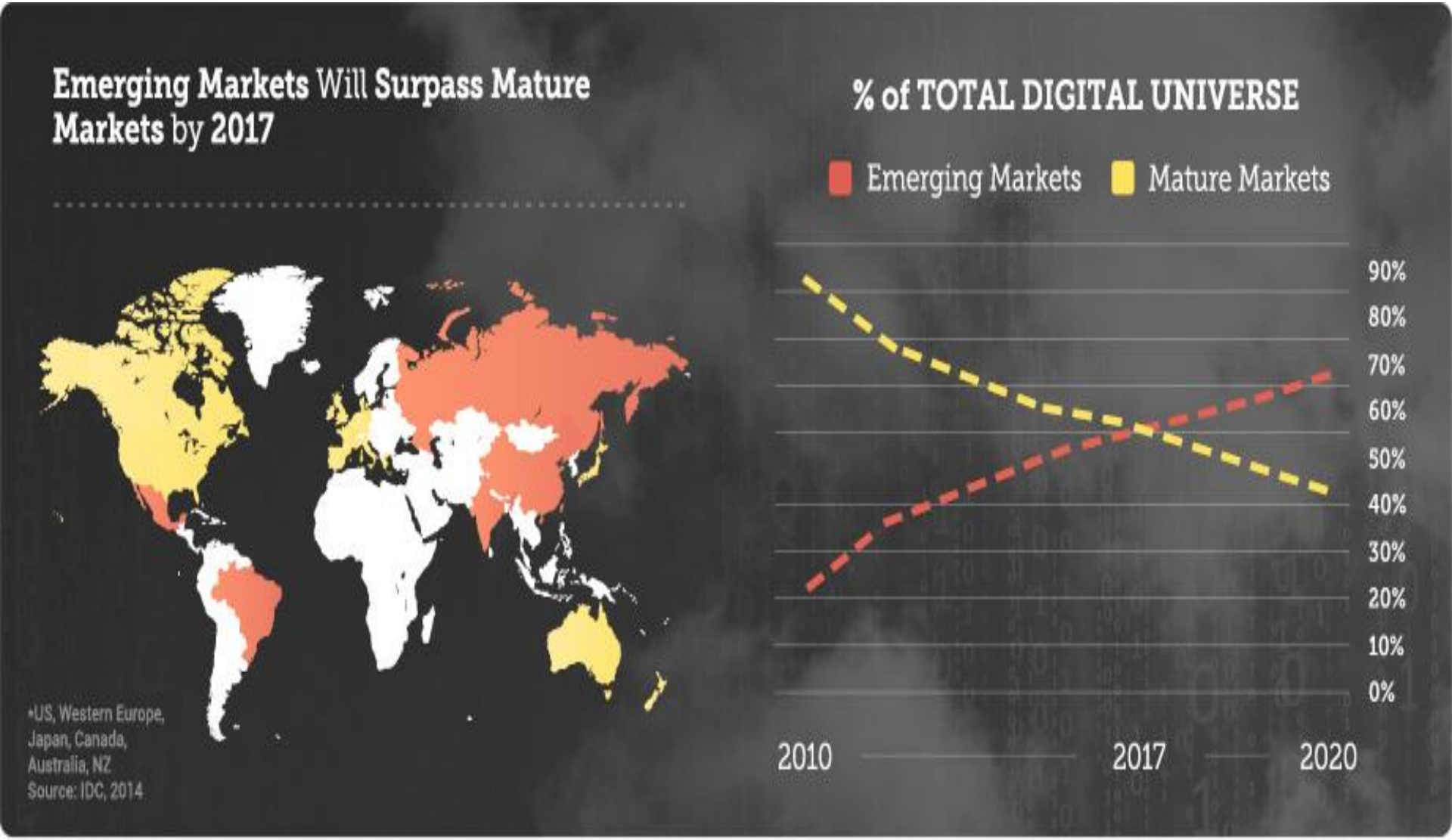
In Domo's 11th edition of Data Never Sleeps, we take the pulse of our digital age, where every click, swipe, and stream fuels an ever-expanding digital universe. These are not just numbers; they are the heartbeat of a world where data reigns supreme.




DATA NEVER SLEEPS AI EDITION
Domo's been monitoring the world's data usage – by the minute – for over a decade now, and the AI phenomenon has caught us all by storm. This recent AI boom has transformed the way businesses operate, students learn and the way people live their everyday lives.
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Learn more at Domo.AI
1 DEEP FAKE OCCURS
467K HOURS ARE SAVED BY EMPLOYEES USING CHATGPT
52 COLLEGE PAPERS ARE FLAGGED FOR AI PLAGIARISM ON TURNITIN
35 NEW CARS WITH HANDS-FREE AUTOMATION ARE SOLD
EVERY MINUTE 01:00 OF THE DAY
$293K IS SPENT ON AI
DALL-E 2 GENERATES 1,389 IMAGES
19,028 IMAGES ARE CREATED USING A TEXT-TO-IMAGE ALGORITHM
185 GALLONS OF H2O COOL THE CHATGPT SUPERCOMPUTERS
CHATGPT IS PROMPTED 6,944 TIMES
7,431 MINUTES OF VIDEO IS GENERATED WITH AI


Annual Size of the Global Datasphere
175 ZB
Zetabytes
180
160
140
120
100
80
60
40
20
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018

The Digital Universe (7th EMC Digital Universe study, IDC, 2014)



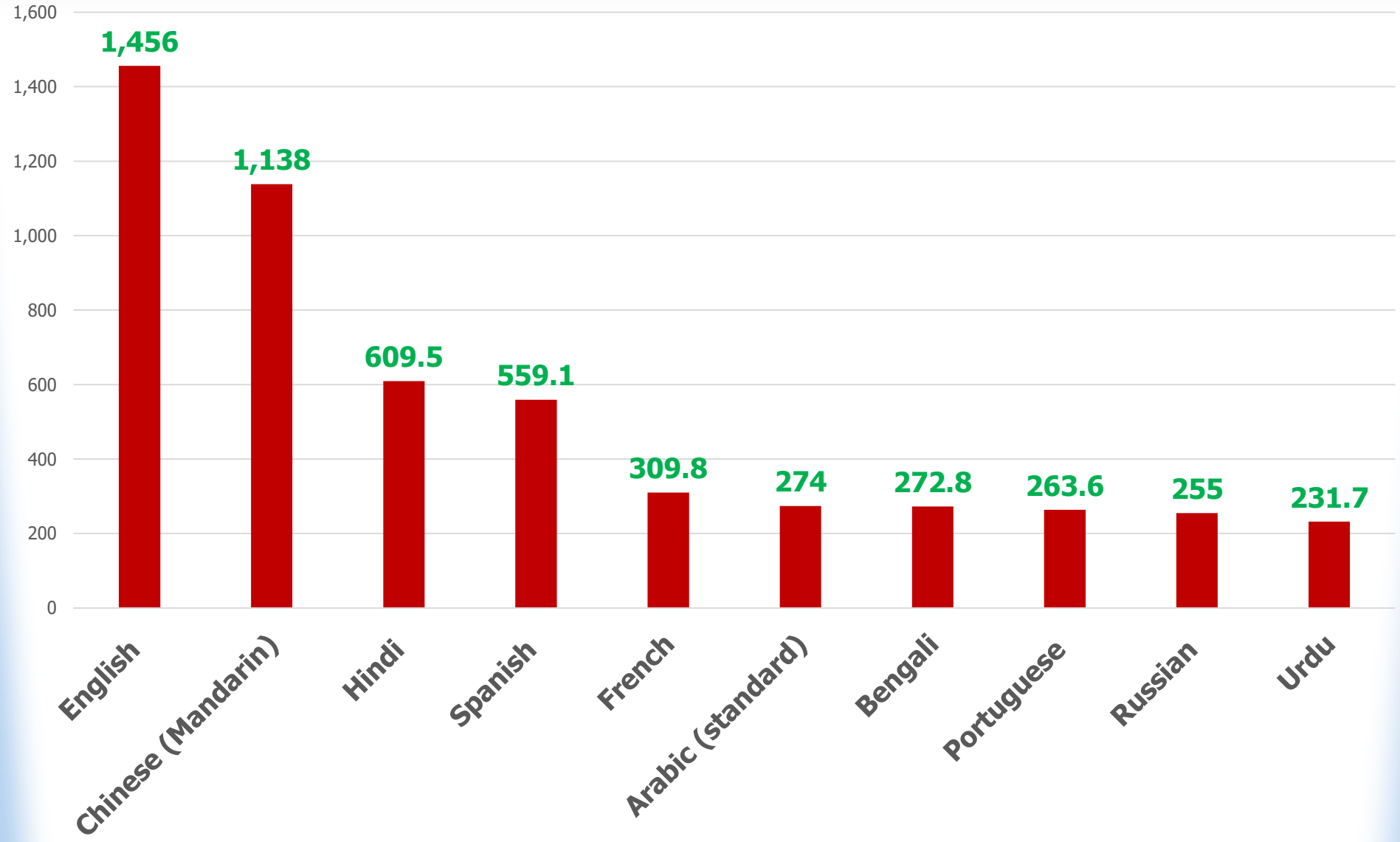
The Digital Universe… The Linguistic Challenge

- Mature markets: USA, Japan, Europe, Canada, Australia, etc.
- Emerging markets: China, Russia, India, Brazil, Mexico, etc.

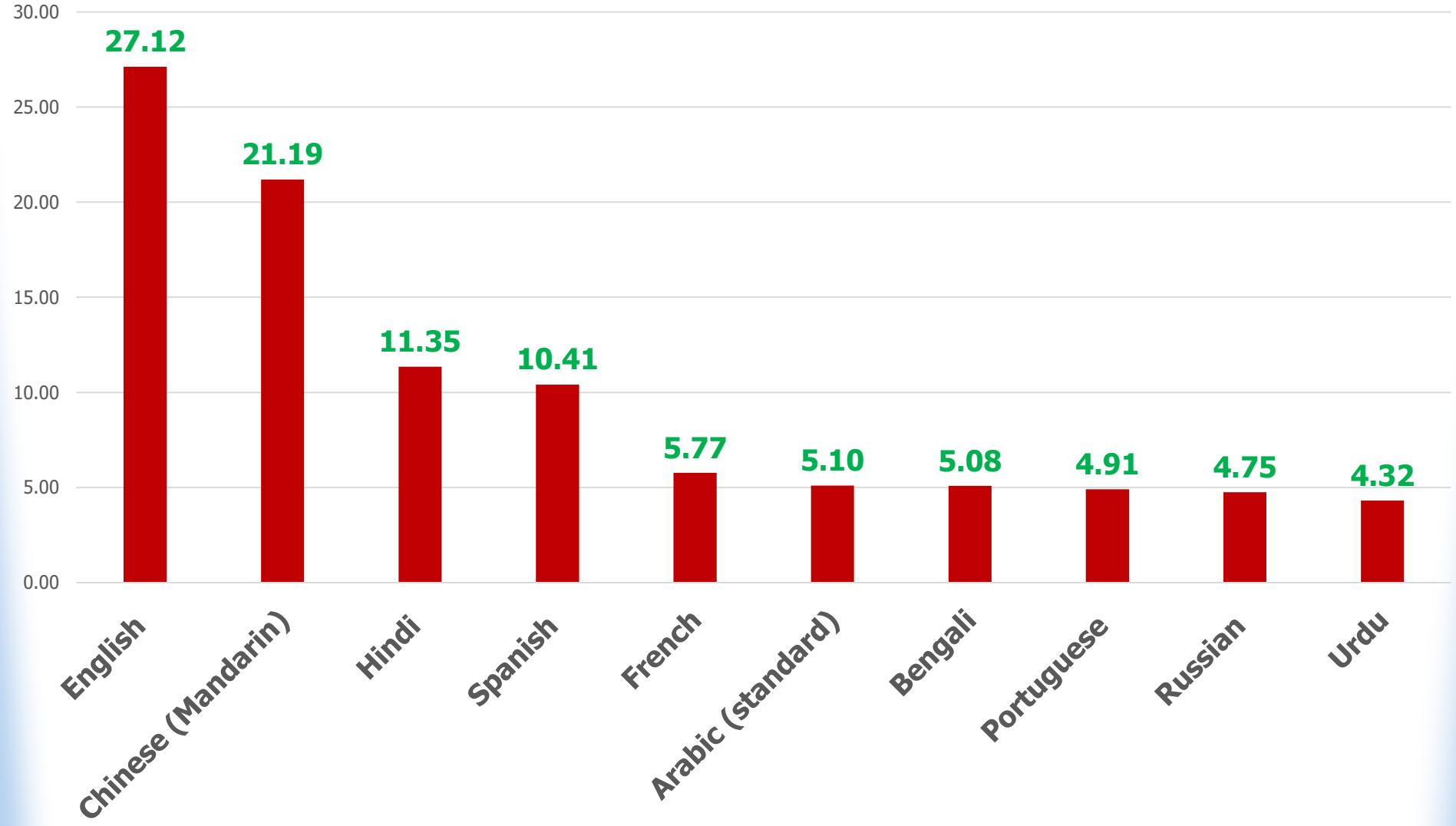
➔ A multilingual digital universe!

(Chinese, Russian, Hindi, Portuguese, Spanish, etc.)

Top 10 Languages on the Internet in 2023 (Millions of users)



Top 10 Languages on Internet in 2023 (% of users)



NLP, a key Solution to the datasphere size and multilingualism !

- Branch of AI at the crossroads of Linguistics, CS, Mathematics, & Statistics.
- **Aim**: to develop IT tools that can understand and generate natural language so as to make the use of technology seamless / intuitive / natural:
  - Answer semantically richer questions by getting information from the Web
  - Get help/advice through conversations with the computer / smart phone (virtual assistants, conversational agents)
  - Automatically generate text summaries
  - Translate documents into different languages
  - Etc.

The Commercial Dimension

Many interesting things in the industry.





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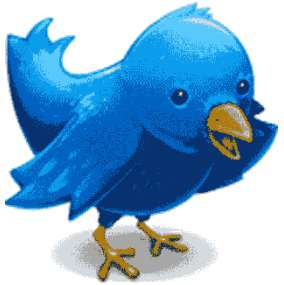
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Applications

- Let us take a quick look at 3 real-world applications:
  - Automatic Translation (Machine Translation)
  - Question answering
  - Text analytics
- And one with an LLM-based conversational agent:
  - ChatGPT

Machine Translation

- Example of Arabic web page and its online translation to English
  - https://www.echoroukonline.com/
- Example of English Web page and its online translation to Arabic
  - Go to translate.google.com
  - And type (e.g.) the URL

    https://www.bbc.com/news
  - Translate (for instance) to Arabic
- Impressive evolution during the last 2 decades! (have a look at slides 22 to 28)

Is MT Poor? (example of translation by SYSTRAN in 2005)

Click the Refresh button, or try again later. If you typed the page address in the Address bar, make sure that it is spelled correctly. To check your connection settings, click the Tools menu, and then click Internet Options. On the Connections tab, click Settings. The settings should match those provided by your local area network (LAN) administrator or Internet service provider (ISP). If your Network Administrator has enabled it, Microsoft Windows can examine your network and automatically discover network connection settings. If you would like Windows to try and discover them, click Detect Network Settings Some sites require 128-bit connection security. Click the Help menu and then click About Internet Explorer to determine what strength security you have installed. If you are trying to reach a secure site, make sure your Security settings can support it. Click the Tools menu, and then click Internet Options. On the Advanced tab, scroll to the Security section and check settings for SSL 2.0, SSL 3.0, TLS 1.0, PCT 1.0. Click the Back button to try another link.

Is MT Poor?
(Systran: Eng. to Fr. in 2005)

Cliquetez le bouton de régénération, ou l'essai encore plus tard. Si vous dactylographiiez l'adresse de page dans la barre d'adresse, assurez-vous qu'elle est orthographiée correctement. Pour vérifier

vos arrangements de raccordement, cliquetez le menu d'outils, et puis cliquetez les options d'Internet. Sur l'étiquette de raccordements, cliquetez les arrangements. Les arrangements devraient assortir ceux fournis par votre Internet Service Provider d'administrateur du réseau local (LAN) ou (ISP). Si votre administrateur de réseau l'a permise, Microsoft Windows peut examiner votre réseau et

automatiquement découvrir des arrangements de raccordement de réseau. Si vous voudriez que Windows les juge et découvre, le déclic détectent des arrangements de réseau quelques emplacements pour exiger la sécurité du raccordement 128-bit. Cliquetez le menu d'aide et puis cliquetez au sujet de l'Internet Explorer pour déterminer quelle sécurité de force vous avez installée. Si vous essayez d'atteindre un emplacement bloqué, assurez-vous que vos arrangements de sécurité peuvent le soutenir. Cliquetez le menu d'outils, et puis cliquetez les options d'Internet. Sur l'étiquette avançée, faites défiler à la section de sécurité et examinez les arrangements pour assurer SSL 2.0, SSL 3.0, TLS 1.0, PCT 1.0.

Cliquetez le bouton arrière pour essayer un autre lien.

Google Translate 18/09/2017

- Cliquez sur le bouton Actualiser ou essayez à nouveau plus tard. Si vous avez tapé l'adresse de la page dans la barre d'adresse, assurez-vous qu'elle est correctement écrite. Pour vérifier vos paramètres de connexion, cliquez sur le menu Outils, puis sur Options Internet. Dans l'onglet Connexions, cliquez sur Paramètres. Les paramètres doivent correspondre à ceux fournis par votre administrateur de réseau local (LAN) ou fournisseur de services Internet (FAI). Si votre administrateur réseau l'a activé, Microsoft Windows peut examiner votre réseau et découvrir automatiquement les paramètres de connexion réseau. Si vous souhaitez que Windows essaye de les découvrir, cliquez sur Détecter les paramètres réseau. Certains sites nécessitent une sécurité de connexion 128 bits. Cliquez sur le menu Aide, puis cliquez sur À propos d'Internet Explorer pour déterminer la sécurité de sécurité que vous avez installée. Si vous essayez de rejoindre un site sécurisé, assurez-vous que vos paramètres de sécurité peuvent le supporter. Cliquez sur le menu Outils, puis cliquez sur Options Internet. Dans l'onglet Avancé, faites défiler jusqu'à la section Sécurité et vérifiez les paramètres pour SSL 2.0, SSL 3.0, TLS 1.0, PCT 1.0. Cliquez sur le bouton Précédent pour essayer un autre lien.

Google Translate 04/10/2024

Cliquez sur le bouton Actualiser ou réessayez plus tard. Si vous avez saisi l'adresse de la page dans la barre d'adresse, assurez-vous qu'elle est correctement orthographiée. Pour vérifier vos paramètres de connexion, cliquez sur le menu Outils, puis cliquez sur Options Internet. Sous l'onglet Connexions, cliquez sur Paramètres. Les paramètres doivent correspondre à ceux fournis par l'administrateur de votre réseau local (LAN) ou votre fournisseur de services Internet (FAI). Si votre administrateur réseau l'a activé, Microsoft Windows peut examiner votre réseau et découvrir automatiquement les paramètres de connexion réseau. Si vous souhaitez que Windows essaie de les découvrir, cliquez sur Détecter les paramètres réseau. Certains sites nécessitent une sécurité de connexion 128 bits. Cliquez sur le menu Aide, puis cliquez sur À propos d'Internet Explorer pour déterminer le niveau de sécurité que vous avez installé. Si vous essayez d'accéder à un site sécurisé, assurez-vous que vos paramètres de sécurité peuvent le prendre en charge. Cliquez sur le menu Outils, puis cliquez sur Options Internet. Dans l'onglet Avancé, faites défiler jusqu'à la section Sécurité et vérifiez les paramètres pour SSL 2.0, SSL 3.0, TLS 1.0, PCT 1.0. Cliquez sur le bouton Retour pour essayer un autre lien.

Is AMT poor? (translation by a well known AMT software - in 2005)

اضغط الينعش الزّرّ أو يحاول ثانية فيما بعد .إذا طبعت عنوان الصّفحة في بار العنوان, تأكّد أنّه تُتَهَجَّى بطريقة سليمة .لفحص أماكن علاقتك, طقطقة قائمة الأدوات ثمّ طقطقة خيارات الإنترنت . على بطاقة العلاقات, يطقطق الإطارات . ينبغي أن تتلاءم الأماكن هؤلاء إذا بشبكتك المحلّيّة (لان) مدير أو شركة خدمات الإنترنت (آي إس بي) . إذا مسئول شبكتك قد مكّن هو, نوافذ مايكروسوفت يمكن أن تفحص شبكتك و تلقائيًّا تكتشف إطارات توصيل الشّبكة .إذا تريد النّوافذ أن تحاول و تكتشفهم, النّقرة تكتشف أماكن الشّبكة بعض المواقع تتطلّب أمن العلاقة ذو ال128 قطعة . طقطق قائمة المساعدة ثمّ اضغط عن إنترنت إكسبلورار للتحديد ماذا أمن القوّة الذي قد ركّبته . إذا تحاول الوصول إلى موقع آمن, التّأكّد أماكنك الأمنيّة يمكن أن تسانده . طقطق قائمة الأدوات ثمّ طقطق خيارات الإنترنت . على البطاقة المتقدّمة, استعرض إلى القسم الأمنيّ و أوضاع الشّيك لإس إس إل 2.0, إس إس إل 3.0, تي إل إس 1.0, بي سي تي 1.0 . طقطق الزّرّ الخلفيّ لمحاولة رابطة أخرى . لا يمكن أن يجد الخادم أو مستكشف دي إن إس إرورينتيرنيت

Google Translate 18/09/2017

- نقر على الزر تحديث، أو أعد المحاولة لاحقا. إذا قمت بكتابة عنوان الصفحة في شريط العنوان، فتأكد من أنه مكتوب بشكل صحيح. للتحقق من إعدادات الاتصال، انقر فوق القائمة أدوات، ثم انقر فوق خيارات إنترنت. في علامة التبويب اتصالات، انقر فوق إعدادات. يجب أن تتطابق الإعدادات مع تلك التي يوفرها مسؤول الشبكة المحلية (لان) أو موفر خدمة الإنترنت (إيسب). إذا كان مسؤول الشبكة لديك قد مكنها، يمكن ميكروسوفت ويندوز فحص الشبكة الخاصة بك واكتشاف إعدادات اتصال الشبكة تلقائيا. إذا كنت ترغب ويندوز في محاولة واكتشافها، انقر فوق كشف إعدادات الشبكة تتطلب بعض المواقع أمان اتصال 128-بت. انقر فوق القائمة تعليمات ثم انقر فوق حول إنترنيت إكسبلورر لتحديد أمان القوة التي قمت بتثبيتها. إذا كنت تحاول الوصول إلى موقع آمن، فتأكد من أن إعدادات الأمان يمكن أن تدعمه. انقر فوق القائمة أدوات ثم انقر فوق خيارات إنترنت. في علامة التبويب خيارات متقدمة، انتقل إلى قسم الأمان وتحقق من إعدادات سل 2.0 و سل 3.0 و تلس 1.0 و يكت 1.0. انقر فوق الخلف زر لمحاولة ارتباط آخر.

Google Translate 05/10/2024

- انقر على الزر "تحديث"، أو حاول مرة أخرى لاحقًا. إذا قمت بكتابة عنوان الصفحة في شريط العناوين، فتأكد من كتابته بشكل صحيح. للتحقق من إعدادات الاتصال، انقر فوق القائمة "أدوات"، ثم انقر فوق "خيارات الإنترنت". في علامة التبويب اتصالات، انقر فوق الإعدادات. يجب أن تتطابق الإعدادات مع تلك التي يوفرها مسؤول الشبكة المحلية (LAN) أو موفر خدمة الإنترنت .(ISP) إذا قام مسؤول الشبكة لديك بتمكين هذه الميزة، فيمكن لنظام التشغيل Microsoft Windows فحص شبكتك واكتشاف إعدادات اتصال الشبكة تلقائيًا. إذا كنت تريد أن يحاول Windowsاكتشافها، فانقر فوق اكتشاف إعدادات الشبكة. تتطلب بعض المواقع أمان اتصال 128 بت. انقر فوق القائمة "تعليمات" ثم انقر فوق "حول Internet Explorer"لتحديد مدى قوة الأمان التي قمت بتثبيتها. إذا كنت تحاول الوصول إلى موقع آمن، فتأكد من أن إعدادات الأمان لديك تدعم ذلك. انقر فوق القائمة "أدوات"، ثم انقر فوق "خيارات الإنترنت". في علامة التبويب خيارات متقدمة، قم بالتمرير إلى قسم الأمان وتحقق من إعدادات SSL 2.0 و SSL 3.0و TLS 1.0 و PCT 1.0. انقر على زر الرجوع لتجربة رابط آخر .

Web Question-Answering

- What's the population of Timimoun

- ChatGPT (conversational Agent)
  - كيف قام الشعب الجزائري بالثورة

- Answer by ChatGPT (05/10/2024)

- الثورة الجزائرية ضد الاستعمار الفرنسي بدأت في 1 نوفمبر 1954، وكانت نتيجة لعدة عوامل تاريخية وسياسية واقتصادية. إليك بعض النقاط الرئيسية حول كيفية قيام الشعب الجزائري بالثورة:
- **الأسباب التاريخية**: عانت الجزائر من الاستعمار الفرنسي منذ عام 1830، حيث تم استغلال الموارد والثروات، وتم تطبيق سياسات تمييزية ضد الجزائريين.
- **الوعي الوطني**: في بداية القرن العشرين، بدأ الوعي الوطني يزداد بين الجزائريين. نشأت حركات سياسية وثقافية تدعو إلى الاستقلال والحقوق المدنية.
- **العدوان والظلم**: تزايدت الأعمال القمعية ضد الشعب الجزائري، مما أدى إلى استياء واسع النطاق. العديد من الجزائريين كانوا يعيشون في ظروف اقتصادية صعبة وحرمان من الحقوق.
- **التنظيم**: في عام 1954، تأسست جبهة التحرير الوطني ((FLNكحركة سياسية وعسكرية تهدف إلى تحقيق الاستقلال. كانت الجبهة منظمة بشكل جيد ولديها دعم شعبي.
- **اندلاع الثورة**: في ليلة 1 نوفمبر 1954، شن مقاتلو جبهة التحرير الوطني هجمات منسقة على مراكز الشرطة والمكاتب الحكومية، مما أطلق شرارة الثورة.
- **الدعم الدولي**: حظيت الثورة بدعم من العديد من الدول والشعوب الأخرى، وأصبحت قضية الجزائر موضوع اهتمام دولي.
- **الحرب الطويلة**: استمرت الثورة حتى عام 1962، وشهدت مواجهات عسكرية شرسة، مع استخدام الطرف الفرنسي لأساليب قمعية، لكن الجبهة نجحت في الحفاظ على مقاومة مستمرة.
- **الاستقلال**: في عام 1962، تم التوصل إلى اتفاق إيفيان، الذي أدى إلى إعلان استقلال الجزائر في 5 يوليو من نفس العام.
- تُعتبر الثورة الجزائرية واحدة من أبرز الحركات التحررية في القرن العشرين، وقد تركت أثراً عميقاً على الهوية الوطنية الجزائرية.

Text Analytics

- Deriving meaningful insights from unstructured text data.
- Various techniques: NLP, statistical analysis, and machine learning.
- Involves transforming text into a structured format that can be analyzed for patterns, trends, and sentiments.
- Helps organizations to:
  - better understand their customers,
  - identify areas for improvement, and
  - make informed business decisions based on real data.

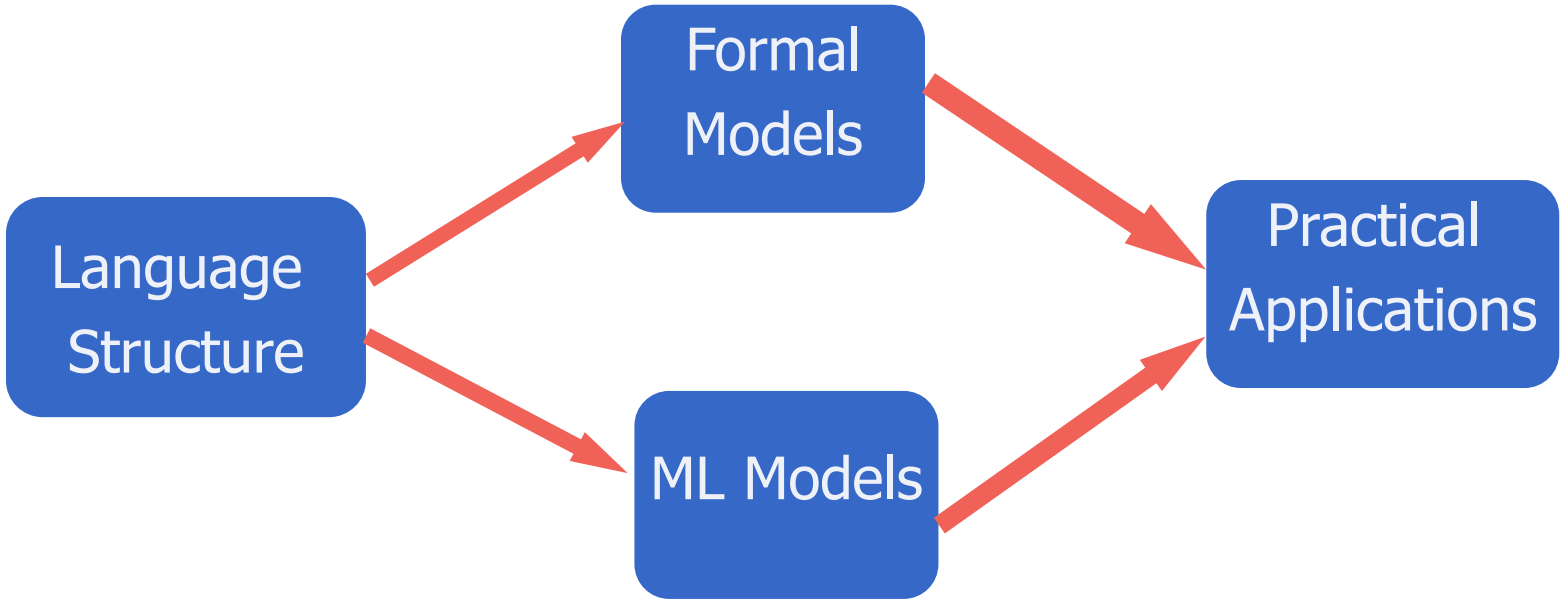
➔ Ultimately leading to enhanced customer satisfaction and loyalty.

Text Analytics

- Data-mining of weblogs, microblogs, discussion forums, message boards, user groups, and other forms of user-generated media (Sentiment Analysis)
- Information for product marketing
- Monitoring political opinions
- Social media analytics
- Tracking the topics of the day (what is "hot", what are the current/discussed topics, etc.)

How are NLP applications developed?

- These applications exploit the underlying structure of human language
- Sometimes in sophisticated ways; other times trivial.
- Sometimes written by experts (rules) or, more recently, through trained models.



NLP's practical applications (Schwartz, Stony Brook U.)

- Machine translation
- Sentiment Analysis
- Automatic speech recognition
  - Personalized assistants
  - Auto customer service
- Information Retrieval
  - Web Search
  - Question Answering
- Computational Social Science
- Growing day by day

How?

- Machine learning:
  - Logistic regression
  - Probabilistic modelling
  - Recurrent Neural Nets
  - Transformers
- Algorithms, e.g.:
  - Graph analytics
  - Dynamic programming
- Data science
  - Hypothesis testing

NLP Applications?

- **Question**: What makes an application an NLP application (rather than just software)?
  - Example: the wc (word count) command on Unix, is this an example of an NLP application?
  - Answer: It must be an application that requires the use of knowledge of the structure of the human language being processed.

NLP Applications?

- Counting words (wc)?
  - When one counts words: Yes
    - To count words, you need to know what a word is ➔ knowledge of the language.
      - NB: *wc* on Unix compiled for English will not work for Chinese!
  - When it counts lines and bytes: No
    - Lines and bytes are programming artifacts, not linguistic entities.
- This is still very simplistic. Developing Conversational Agents (e.g. HAL) requires much deeper/richer modeling of a language!

Classes of NLP Applications

- Small
  - Spelling correction
  - Segmentation
  - Etc.
- Medium
  - Word sense disambiguation
  - Named Entity Recognition
  - Information Retrieval
- Big
  - Questions-Answering Systems
  - Conversational Agents
  - Machine Translation

The big NLP Applications

- These types of applications require a large amount and a good variety of knowledge
- Consider for instance the exchange with HAL the artificial agent of "2001: *A Space Odyssey*"
- *Dave Bowman: Open the pod bay doors, HAL.*
- *HAL: I'm sorry Dave, I'm afraid I can't do that.*

(Stanley Kubrick and Arthur C. Clarke)

What is required?

- Speech analysis and generation
- Understanding the words of the language
  - their meaning(s)
- How groups of words are "concatenated" together
  - what these concatenations mean
- Dialogues (rules):
  - It is always polite to respond, whatever the purpose.
  - It is always polite to pretend to want to be cooperative (I'm afraid, I won't be able to...)

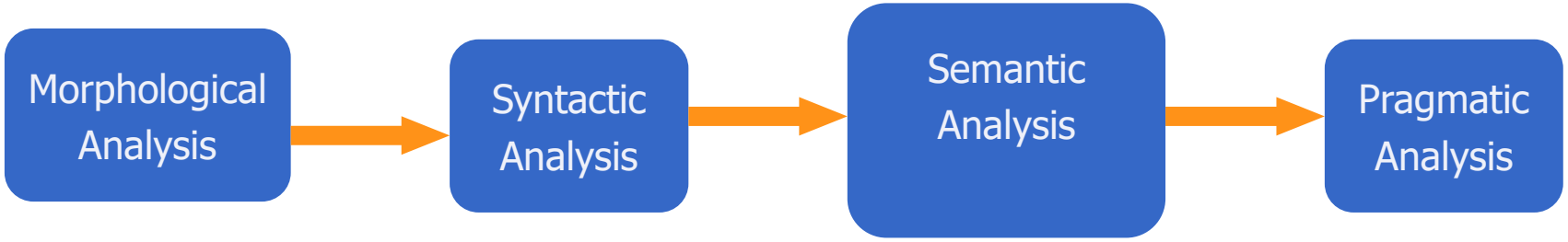
Types of Knowledge of Language

- **Phonetics and Phonology**: knowledge about linguistic sounds.
- **Morphology:** knowledge of the meaningful components of words.
- **Syntax:** knowledge of the structural relationships between words.
- **(Lexical & compositional) Semantics :** knowledge of meaning.
- **Pragmatics:** knowledge of the relationship of meaning to the goals and intentions of the speaker.
- **Discourse:** knowledge about linguistic units larger than a single utterance. E.g. How many states were in the US *that year*? (case of **coreference resolution**)

Types of Knowledge of Language

- Each type of knowledge is associated with a set of processes that use it
- Interfaces are defined such that they allow the different levels to communicate.

  ➔ A pipeline architecture



Ambiguity

- Ambiguity is a fundamental problem in computational linguistics.
- Resolving ambiguities is crucial and challenging.
- Examples:
  - If the **baby** does not thrive on **cow**'s **milk**, boil **it**!
  - I made her duck

Ambiguity

- At least five different meanings this sentence could have, each of which exemplifies an ambiguity at some level:
  1. I cooked waterfowl for her.
  2. I cooked waterfowl belonging to her.
  3. I created the (wooden/ceramic?) duck she owns.
  4. I caused her to quickly lower her head or body.
  5. I waved my magic wand and turned her into a waterfowl.

Ambiguity is complex!

- I caused her to quickly lower her head or body:
  - **Lexical Category**: "duck" can be a noun or a verb
- I cooked waterfowl belonging to her:
  - **Lexical Category**: "her" can be possessive pronoun ("of her") or dative pronoun ("for her")
- I created the (wooden/ceramic?) duck she owns
  - **Lexical Semantics**: "make" can mean "to make" or "to cook", and many other meanings…

Ambiguity is complex!

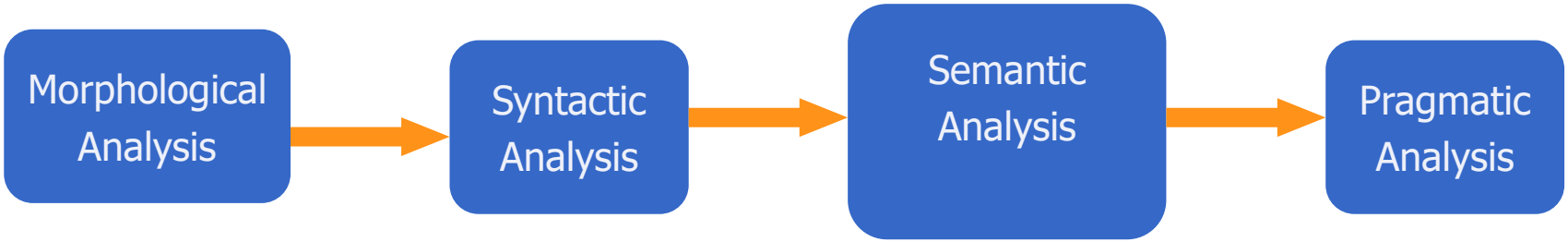
- **Grammar**: "Make" can be:
  - **Transitive**: the verb takes a noun as direct object.
    - *I cooked [a duck belonging to her].*
  - **Ditransitive**: the verb takes 2 nouns as objects.
    - *I made [her] (into) [a duck] .*
  - **Action-transitive**: the verb takes a direct object and another verb.
    - *I caused [her] [to move her body]* .

Ambiguity is complex!

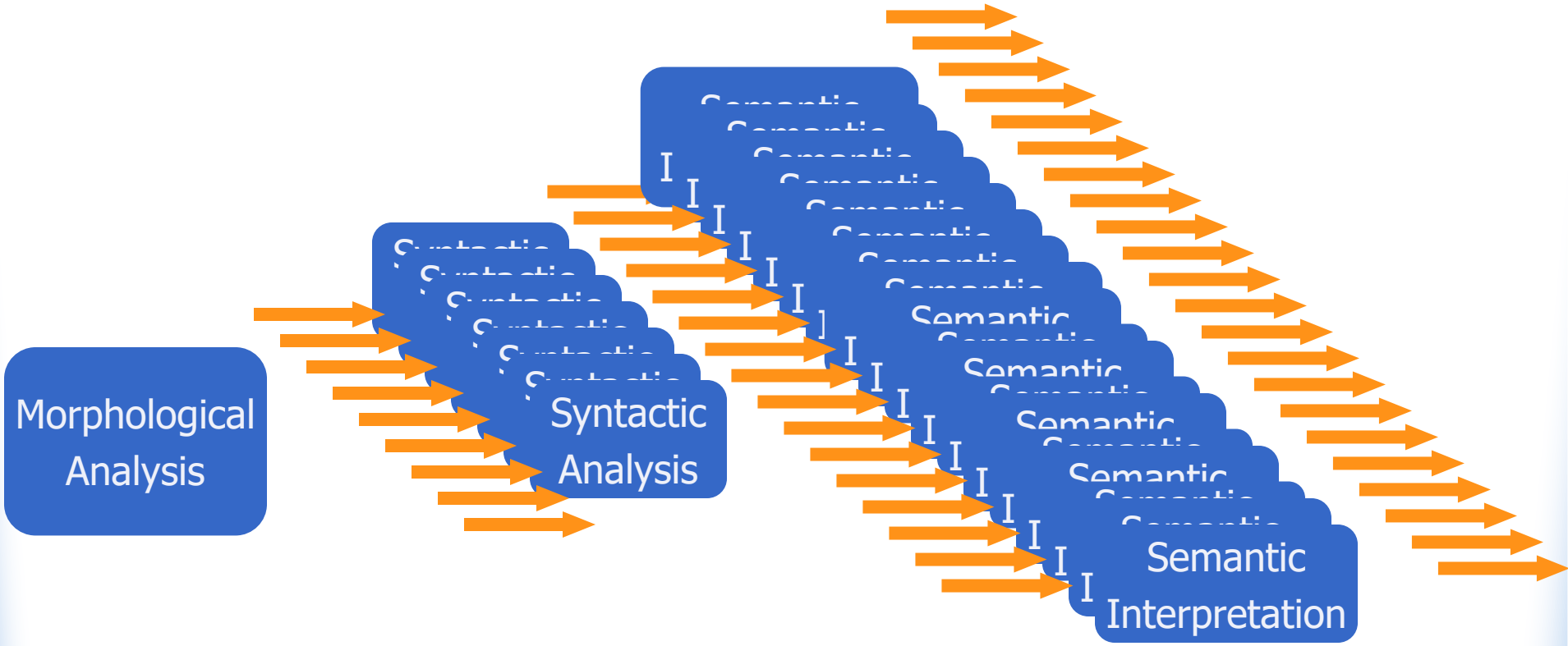
- **Adding Phonetics to the above!**
  - I mate or duck
  - I'm eight or duck
  - Eye maid; her duck
  - Aye mate, her duck
  - I maid her duck (*)
  - I'm aid her duck (*)
  - I mate her duck
  - I'm ate her duck (*)
  - I'm ate or duck (*)

Problem

- Recall the pipeline architecture



It is actually more like…



Handling Ambiguity

- Various possible approaches:
  1. Strongly linked interactions between the different processing levels; knowledge acquired from other levels can help choose between the different levels of ambiguity.

  2. Pipelined processing that ignores ambiguity when it arises and hopes that the next levels can eliminate incorrect structures/analyses.

Handling Ambiguity

3. Probabilistic approaches where the choice is made on the most plausible case
   - Or move on to the next level with the n best choices

4. Approach based on Machine Learning: train models on large corpora.

5. Do nothing; maybe this won't have an impact

Models and Algorihms

- By **models** we mean formalisms used to capture the different types of linguistic knowledge that we need.
  - Concept of **language model**.
- **Algorithms** are then used to manipulate the language knowledge representations necessary to solve the problem.

Models

In NLP different types of models can be developed and used for different purposes, whichever is most appropriate:

- State machines (FSM, FST, Weighted, …)
- Production rules (Regular Grammars, CFG, Feature-Augmented Grammars, …)
- Logical formalisms (FOL, and other logics)
- Probabilistic models
- Vector-Space models (TF-IDF, Bag-of-Words, Word Embeddings, …)

Algorithms

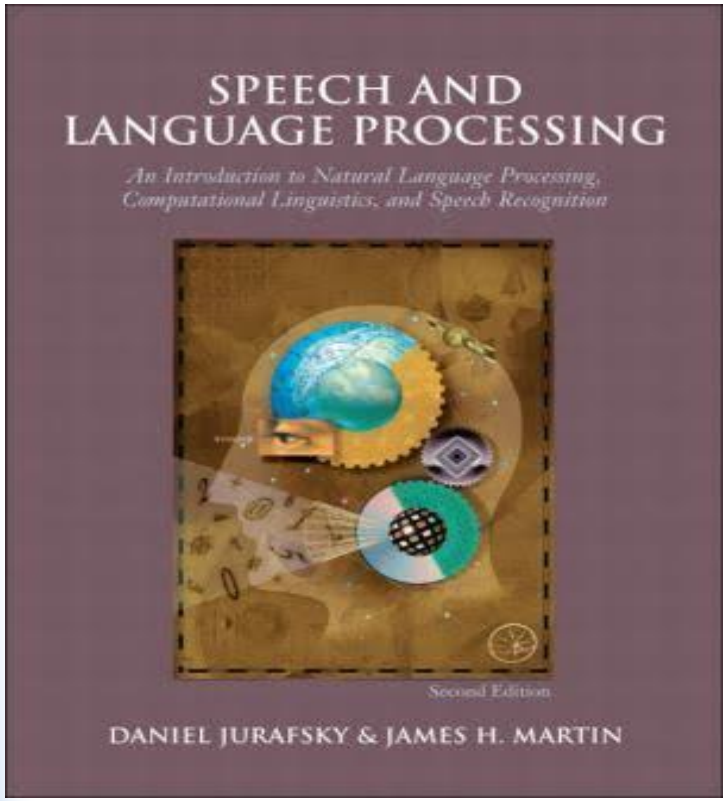
- Algorithms are needed to handle the developed models:
  - **Space search algorithms**, e.g. DFS, BFS, A*, dynamic programming.
  - **Machine Learning algorithms** such as
    - Various types of **Classifiers** (DT, SVM, Logistic regression, etc.)**,**
    - **Sequence Models** (HMMs, Maximum Entropy Markov Models, …),
    - Sequence-to-Sequence, Transformers, ...

Chapter References

- **Data Never Sleeps AI Edition**: https://finance.yahoo.com/news/domo-releases-data-never-sleeps-130000727.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAGZQUL1dgcufmgSZPt3Cg-ferwW5EQq3HUhtzEs7TOzCSFYDVoPsBQEkyQfts2m3-w3GI7tmTx67qDLaqe_Nxb8b87I7XWQhOm5ru4mIKN4TkhBbjQD0qJZ9ndBRchj61dYf5YtzzvIwrHkSllN3GGIUphztAVdL3eirhTg0BVM
- Edge Delta, *Breaking Down The Numbers: How Much Data Does The World Create Daily in 2024*? 11/03/2024 https://edgedelta.com/company/blog/how-much-data-is-created-per-day#:~:text=Google%20manages%20approximately%2020%20PB%20every%20day.,-(Skill%2DLync%2C&text=The%20search%20engine%20processes%20approximately,Google%20searches
- Slide 16: https://www.statista.com/statistics/266808/the-most-spoken-languages-worldwide/. Slide 17 calculations based on same reference.
- Slide 34 from Andrew Schwartz, Stony Brook University

Textbook

Jurafsky, D. and Martin, J.H. (2024) *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models* (3rd ed. draft), Prentice Hall.
https://web.stanford.edu/~jurafsky/slp3/ed3bookaug20_2024.pdf



The draft version of the 3rd edition will be used; it has been updated to include more recent topics such as Vector Semantics and Embeddings, ANNs and Deep Learning for NLP, Transformers, etc.

This course **slides** are largely based on this textbook and its authors' slides with light customisations whenever seen fit.

This chapter: Slides 8-17 & 21-32 independent researching. (Refs. given)